



ROBOGAMBIT

Smart Moves. Precision Play.

AI Chess Robot • Computer Vision • ROS 2 • PySide6 GUI

Intelligent Robotics Course — Final Project

- **What is RoboGambit?**

A robot that plays chess on a real board.

A camera watches the game. Computer vision detects your move. Stockfish thinks. A 5-DOF robotic arm physically picks up and places the piece — while a polished GUI mirrors everything in real time.



SEE

Camera + OpenCV



THINK

Stockfish AI



MOVE

Robotic Arm

● System Architecture

Modular pipeline. Loosely coupled. ROS 2 between layers.



Communication via ROS 2 topics: `/robogambit/move` → `/nano_serial`

- 🔍 Each layer can be swapped or tested independently
- 🔄 Standard ROS topics — easy to extend with feedback loops
- 🖥️ Same GUI works in fake-mode (no hardware) or real-mode

● Computer Vision — Detecting Moves on a Real Board



64-square pixel calibration



Board Calibration

Click 4 corners → maps every chess square to a pixel polygon (sqdict.json).



Frame Differencing

Compare before/after frames → finds the 2 changed squares = your move.



Auto-Detect Mode

Motion-then-stillness state machine: just play, no buttons to click.



Live Preview

Camera feed embedded in the GUI — see exactly what the system sees.

● AI Engine & Game Modes



Stockfish Engine

- World-class chess engine integrated via python-chess
- Three difficulty levels: Easy / Medium / Hard
- Skill levels mapped to Stockfish skill setting
- Smart move hints — toggleable per player
- Full chess rule support: captures, castling, en passant, promotion
- Move legality validation — invalid moves are gracefully rejected

Four Game Modes



Player vs AI (Robot)

Real board, real arm



Player vs AI

On-screen vs Stockfish



Player vs Player

Two humans + hints

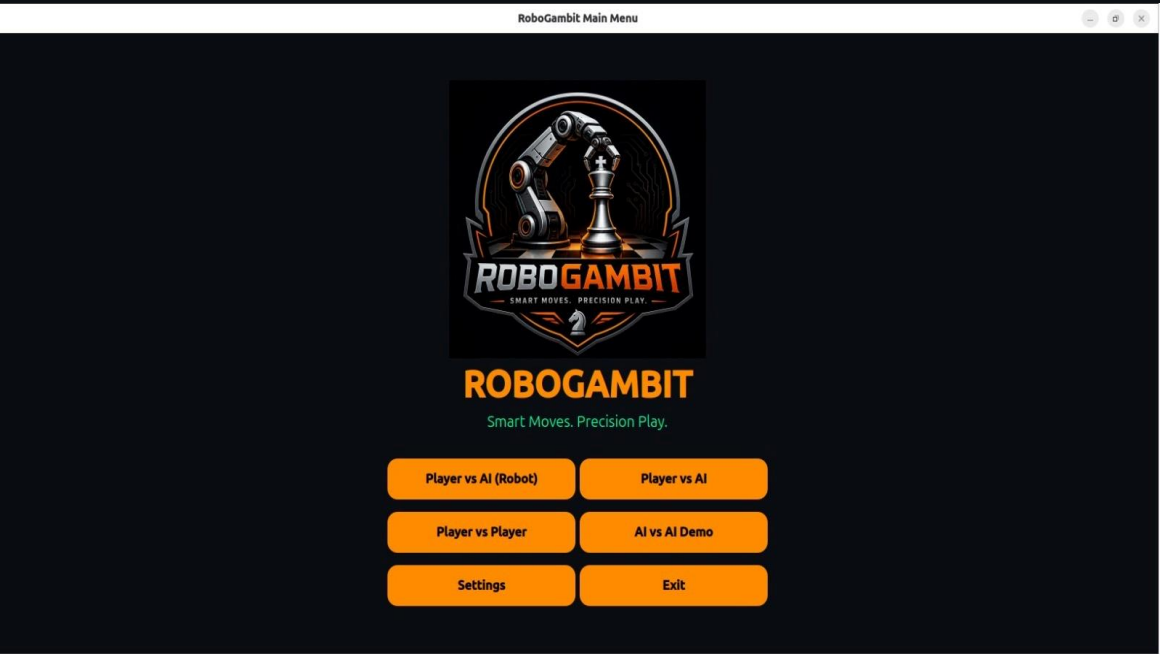


AI vs AI Demo

Watch Stockfish self-play

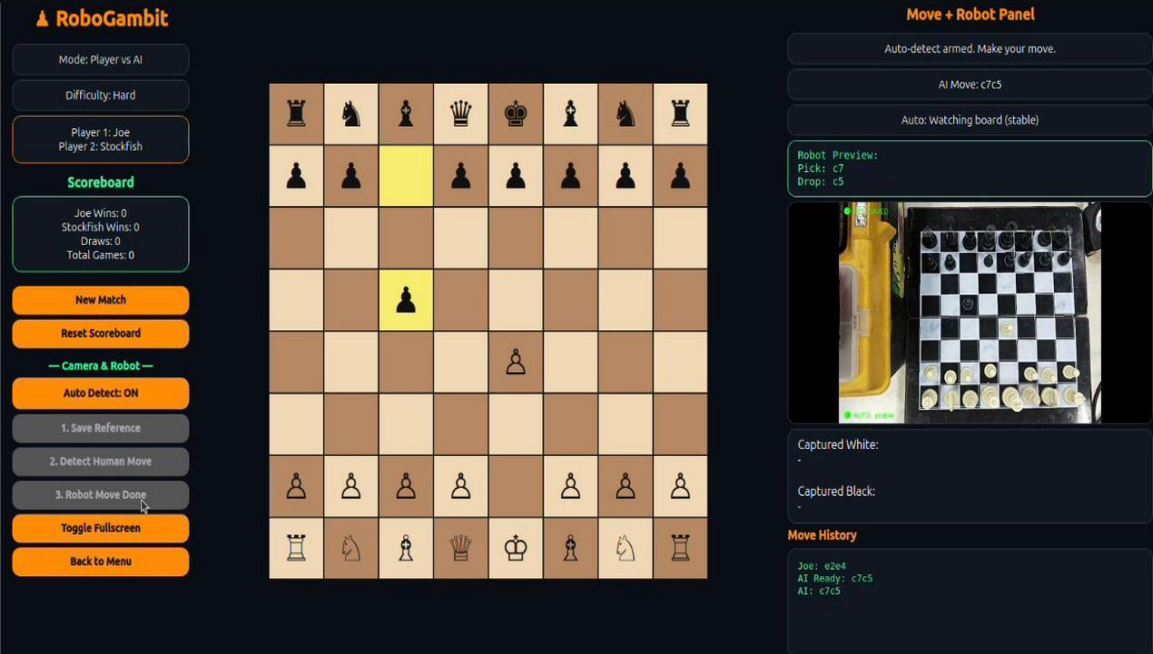
● GUI Showcase

Built with PySide6 — dark theme, fullscreen, and live camera feed.



Main Menu

Game-mode selection with branded logo



In-Game View

Board + live camera + history + robot status

🎯 Capture highlighting

📜 Move history

🏆 Scoreboard

💡 Per-player hints

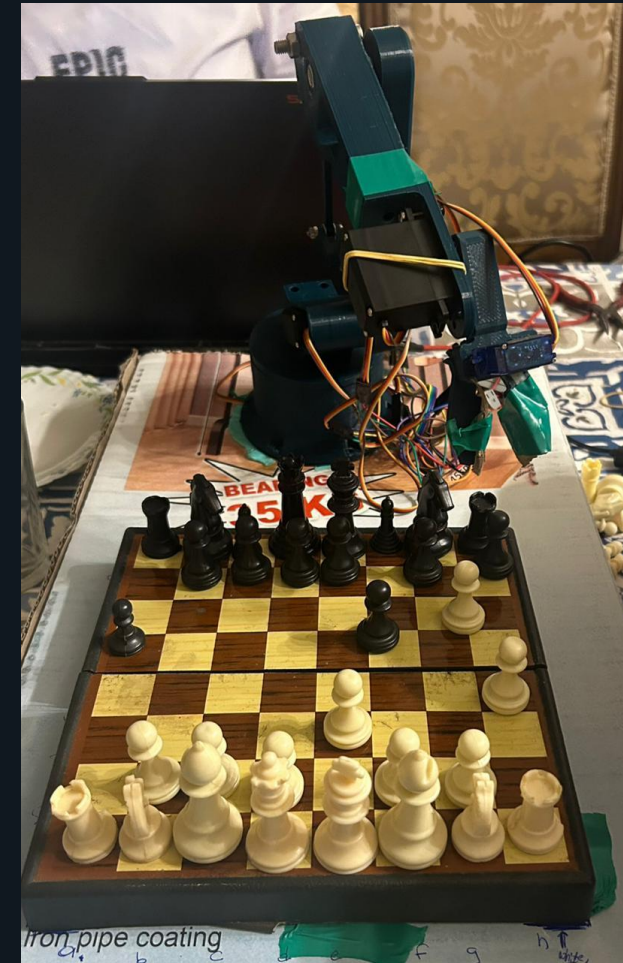
🖥️ Fullscreen mode

● Hardware & Arduino Firmware



The Arm — 5 DOF

- 5 servos: base, shoulder, elbow, wrist, gripper
- Arduino Nano controller (9600 baud serial)
- External 5V supply for servo power rail
- Smooth interpolated movement with per-servo limits
- Single-command protocol: e.g. "S1 165\n" sets servo 1 to 165°
- Two off-board zones for captured pieces (left / right)



• ROS 2 Control Pipeline

From chess move to servo angles — three nodes, two topics, one robot.

1

GUI Publisher

Publishes UCI move strings

`main.py` → `robot/ros_robot.py`



`/robogambit/move`



`"e2e4"`

2

IK Translator Node

Chess square → servo commands

Calibrated lookup table per square



`/nano_serial`



`"S2 140" . "S1 180"`

3

Serial Bridge Node

Forwards commands to Arduino over USB

Auto-detects Arduino by VID/PID

● The Team

Intelligent Robotics Course — Final Project



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Hardware • ROS 2 • Firmware

SEE. THINK. MOVE.

Live demo coming up next.

 RoboGambit